

▣ Gradient, Divergence and curl.

Definition:

Del/Nabla: The vector differential operator Del, written $\vec{\nabla}$ is defined by,

$$\begin{aligned}\vec{\nabla} &= \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \\ &= \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\end{aligned}$$

The operator $\vec{\nabla}$ is also known as nabla.

The Gradient: Let $\phi(x, y, z)$ be defined and differentiable at each point (x, y, z) in a certain region of space. The gradient of ϕ written $\vec{\nabla}\phi$ or $\text{grad } \phi$ is defined by,

$$\begin{aligned}\vec{\nabla}\phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \\ &= \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}\end{aligned}$$

~~The Curl is~~

Divergence: Let $\vec{V}(x, y, z) = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$ be defined and differentiable at each point (x, y, z) in a certain region of space. Then the divergence of $\vec{V}$, written $\vec{\nabla} \cdot \vec{V}$ or $\text{div } \vec{V}$ is defined by

$$\begin{aligned}\vec{\nabla} \cdot \vec{V} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}) \\ &= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}\end{aligned}$$

Curl: If $\vec{V}(x, y, z)$ is a differentiable vector field then the curl or rotation of $\vec{V}$, written $\vec{\nabla} \times \vec{V}$, curl $\vec{V}$ or ~~rot~~ $\vec{V}$ is defined by,

$$\begin{aligned}\vec{\nabla} \times \vec{V} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}\end{aligned}$$

Solenoidal vector: A solenoidal vector is a vector field with zero divergence, meaning there are no sources or sinks. i.e; $\vec{\nabla} \cdot \vec{V} = 0$.

Irrrotational vector: An irrotational vector is a vector field with zero curl, meaning it has no rotation or circulation. i.e; $\vec{\nabla} \times \vec{V} = 0$.

Gradient:

① If $\phi(x, y, z) = 3x^2y - y^3z^2$ find $\vec{\nabla}\phi$ (or grad ϕ) at the point $(1, -2, 1)$

Solution: Given that,

$$\phi(x, y, z) = 3x^2y - y^3z^2$$

Also know that, $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$

Now,

$$\begin{aligned}\vec{\nabla}\phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (3x^2y - y^3z^2) \\ &= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2) \\ &= 6xy \hat{i} + (3x^2 - 3y^2z^2) \hat{j} - 2y^3z \hat{k}\end{aligned}$$

At the point $(1, -2, 1)$,

$$\begin{aligned}\vec{\nabla}\phi &= 6 \cdot 1 \cdot (-2) \hat{i} + \{3 \cdot 1^2 - 3(-2)^2 \cdot (-1)^2\} \hat{j} - \{2 \cdot (-2)^3 \cdot (-1)\} \hat{k} \\ &= -12 \hat{i} - 9 \hat{j} - 16 \hat{k}\end{aligned}$$

(Ans.)

③ Find $\vec{\nabla}\phi$ if (a) $\phi = \ln|\vec{r}|$; (b) $\phi = \frac{1}{r}$

Solution: We know that,

The position vector of particle, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\text{Then } |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

a. Now, $\phi = \ln|\vec{r}|$

$$= \ln(\sqrt{x^2 + y^2 + z^2})$$

$$= \frac{1}{2} \ln(x^2 + y^2 + z^2)$$

$$\begin{aligned}
 \therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \left(\frac{1}{2} \ln(x^2 + y^2 + z^2) \right) \\
 &= \frac{1}{2} \hat{i} \frac{\partial}{\partial x} \{ \ln(x^2 + y^2 + z^2) \} + \frac{1}{2} \hat{j} \frac{\partial}{\partial y} \{ \ln(x^2 + y^2 + z^2) \} \\
 &\quad + \frac{1}{2} \hat{k} \frac{\partial}{\partial z} \{ \ln(x^2 + y^2 + z^2) \} \\
 &= \frac{1}{2} \cdot \frac{2x}{x^2 + y^2 + z^2} \hat{i} + \frac{1}{2} \cdot \frac{2y}{x^2 + y^2 + z^2} \hat{j} + \frac{1}{2} \cdot \frac{2z}{x^2 + y^2 + z^2} \hat{k} \\
 &= \frac{1}{x^2 + y^2 + z^2} (x\hat{i} + y\hat{j} + z\hat{k}) \\
 &= \frac{\vec{r}}{r^2} \text{ (Ans.)}
 \end{aligned}$$

⑥ Now, $\phi = \frac{1}{r}$

$$\begin{aligned}
 \therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) \\
 &= \hat{i} \frac{\partial}{\partial x} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) + \hat{j} \frac{\partial}{\partial y} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) + \hat{k} \frac{\partial}{\partial z} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) \\
 &= \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2x \right\} \hat{i} + \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2y \right\} \hat{j} + \left\{ -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2z \right\} \hat{k} \\
 &= -(x^2 + y^2 + z^2)^{-3/2} (x\hat{i} + y\hat{j} + z\hat{k}) \\
 &= \frac{-\vec{r}}{r^3} \\
 &\text{ (Ans.)}
 \end{aligned}$$

— 0 —

⑤ Show that $\vec{\nabla} \phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$ where c is a constant.

Solution: Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector to any point $P(x, y, z)$ on the surface.
 $\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ lies in the tangent plane to the surface at P .

Given $\phi(x, y, z) = c$

$$\text{or, } d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = d(c)$$

$$\text{or, } d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = 0$$

$$\text{or, } \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = 0 \quad \text{--- (i)}$$

$$\text{or, } \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0$$

$$\text{or, } \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0 \quad \text{--- (ii)}$$

$$\text{or, } \vec{\nabla} \phi \cdot d\vec{r} = 0$$

So that $\vec{\nabla} \phi$ perpendicular to $d\vec{r}$, and

[Shown]

$$(\hat{i}x + \hat{j}y + \hat{k}z) \cdot (\hat{i}x + \hat{j}y + \hat{k}z) =$$

(Ans.)

⑥ Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.

Solution: Given that,

$$\phi(x, y, z) = x^2y + 2xz = 4$$

$$\therefore \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$= \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \frac{\partial}{\partial x} (x^2y + 2xz) + \hat{j} \frac{\partial}{\partial y} (x^2y + 2xz) + \hat{k} \frac{\partial}{\partial z} (x^2y + 2xz)$$

$$= (2xy + 2z) \hat{i} + x^2 \hat{j} + 2x \hat{k}$$

At the point $(2, -2, 3)$,

$$\vec{\nabla} \phi = \{2 \cdot 2(-2) + 2(3)\} \hat{i} + (2^2) \hat{j} + 2(2) \hat{k}$$

$$= -2\hat{i} + 4\hat{j} + 4\hat{k}$$

$$\therefore \text{A unit normal to the surface} = \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{(-2)^2 + 4^2 + 4^2}}$$

$$= \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{36}}$$

$$= \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{6}$$

$$= \frac{-1}{3} \hat{i} + \frac{2}{3} \hat{j} + \frac{2}{3} \hat{k}$$

(Ans.)

— 0 —

⑦ Find the equation for the tangent plane to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$.

Solution: Given that, $r = x\hat{i} + y\hat{j} + z\hat{k} = (x, y, z)$
 $\phi(x, y, z) = 2xz^2 - 3xy - 4x = 7$ $\nabla \phi = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k} = 0$ $\therefore$

$$\frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k} =$$

$$\frac{\partial}{\partial x}(2xz^2 - 3xy - 4x)\hat{i} + \frac{\partial}{\partial y}(2xz^2 - 3xy - 4x)\hat{j} + \frac{\partial}{\partial z}(2xz^2 - 3xy - 4x)\hat{k} =$$

$$(2z^2 - 3y - 4)\hat{i} + (-3x)\hat{j} + (4xz)\hat{k} =$$

At the point $(1, -1, 2)$ $\nabla \phi =$

$$(2(2)^2 - 3(-1) - 4)\hat{i} + (-3(1))\hat{j} + (4(1)(2))\hat{k} = (5)\hat{i} - 3\hat{j} + 8\hat{k}$$

$$\hat{i} + \hat{j} + \hat{k} =$$

$$\frac{(5)\hat{i} - 3\hat{j} + 8\hat{k}}{\sqrt{5^2 + (-3)^2 + 8^2}} = \frac{(5)\hat{i} - 3\hat{j} + 8\hat{k}}{\sqrt{94}}$$

$$\frac{(5)\hat{i} - 3\hat{j} + 8\hat{k}}{\sqrt{94}} =$$

$$\frac{(5)\hat{i} - 3\hat{j} + 8\hat{k}}{\sqrt{94}} =$$

$$\frac{(5)\hat{i} - 3\hat{j} + 8\hat{k}}{\sqrt{94}} =$$

(Ans)

10. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$.

Solution: Given that,

$$\phi = x^2yz + 4xz^2$$

$$\text{Also know that, } \nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\begin{aligned} \therefore \nabla \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2yz + 4xz^2) \\ &= \hat{i} \frac{\partial}{\partial x} (x^2yz + 4xz^2) + \hat{j} \frac{\partial}{\partial y} (x^2yz + 4xz^2) \\ &\quad + \hat{k} \frac{\partial}{\partial z} (x^2yz + 4xz^2) \\ &= (2xyz + 4z^2) \hat{i} + x^2z \hat{j} + (x^2y + 8xz) \hat{k} \end{aligned}$$

Now, at the point $(1, -2, -1)$

$$\begin{aligned} \nabla \phi &= \{2 \cdot 1 \cdot (-2) \cdot (-1) + 4(-1)^2\} \hat{i} + 1^2 \cdot (-1) \hat{j} + \{1^2 \cdot (-2) + 8 \cdot 1 \cdot (-1)\} \hat{k} \\ &= 8\hat{i} - \hat{j} - 10\hat{k} \end{aligned}$$

Let, the given direction vector,

$$\vec{A} = 2\hat{i} - \hat{j} - 2\hat{k}$$

unit direction vector, $\hat{a} = \frac{\vec{A}}{|\vec{A}|}$

$$= \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{2^2 + (-1)^2 + (-2)^2}}$$

$$\therefore \hat{a} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

The required directional derivative,

$$\begin{aligned} \nabla \phi \cdot \hat{a} &= (8\hat{i} - \hat{j} - 10\hat{k}) \cdot \left(\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k} \right) \\ &= \frac{16}{3} + \frac{1}{3} + \frac{20}{3} \\ &= \frac{37}{3} \text{ (Ans.)} \end{aligned}$$

11. (a) In what direction from the point $(2, 1, -1)$ is the directional derivative of $\phi = x^2 y z^3$ a ~~max~~ maximum?

(b) What is the magnitude of this maximum?

Solution:

(a) Given that,

$$\phi(x, y, z) = x^2 y z^3$$

$$\text{Also know that, } \vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 y z^3) \\ &= \hat{i} \frac{\partial}{\partial x} (x^2 y z^3) + \hat{j} \frac{\partial}{\partial y} (x^2 y z^3) + \hat{k} \frac{\partial}{\partial z} (x^2 y z^3) \\ &= 2x y z^3 \hat{i} + x^2 z^3 \hat{j} + 3x^2 y z^2 \hat{k} \end{aligned}$$

At the point $(2, 1, -1)$

$$\begin{aligned} \vec{\nabla} \phi &= \{ 2 \cdot 2 \cdot 1 \cdot (-1)^3 \hat{i} + (2)^2 \cdot (-1)^3 \hat{j} + 3(2)^2 \cdot 1 \cdot (-1)^2 \hat{k} \} \\ &= -4\hat{i} - 4\hat{j} + 12\hat{k} \quad (\text{Ans.}) \end{aligned}$$

(b) The magnitude of this maximum $\equiv |\vec{\nabla} \phi|$

$$= \sqrt{(-4)^2 + (-4)^2 + (12)^2}$$

$$= \sqrt{176}$$

$$= 4\sqrt{11}$$

(Ans.)

12 Find the angle between the surfaces:
 $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point
 $(2, -1, 2)$.

Solution: Given that,

$$z = x^2 + y^2 - 3$$

$$\Rightarrow x^2 + y^2 - z = 3$$

$$\text{Let, } \phi_1(x, y, z) = x^2 + y^2 + z^2 = 9$$

$$\phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\begin{aligned}\therefore \vec{\nabla} \phi_1 &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi_1 \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2 + z^2) \\ &= 2x \hat{i} + 2y \hat{j} + 2z \hat{k}\end{aligned}$$

At the point $(2, -1, 2)$,

$$\begin{aligned}\vec{\nabla} \phi_1 &= 2(2) \hat{i} + 2(-1) \hat{j} + 2(2) \hat{k} \\ &= 4 \hat{i} - 2 \hat{j} + 4 \hat{k}\end{aligned}$$

Again,

$$\begin{aligned}\vec{\nabla} \phi_2 &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2 - z) \\ &= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 - z) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 - z) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 - z) \\ &= 2x \hat{i} + 2y \hat{j} - \hat{k}\end{aligned}$$

At the point $(2, -1, 2)$

$$\therefore \vec{\phi}_2 = 2(2)\hat{i} + 2(-1)\hat{j} - \hat{k}$$

$$= 4\hat{i} - 2\hat{j} - \hat{k}$$

Let, θ be the angle between ϕ_1 and ϕ_2 ,

$$\text{so, } \vec{\phi}_1 \cdot \vec{\phi}_2 = |\vec{\phi}_1| |\vec{\phi}_2| \cos \theta$$

$$\text{or, } (4\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (4\hat{i} - 2\hat{j} - \hat{k}) = \sqrt{4^2 + (-2)^2 + 4^2} \cdot \sqrt{4^2 + (-2)^2 + (-1)^2}$$

$$\cos \theta$$

$$\text{or, } 16 + 4 - 4 = 6 \cdot \sqrt{21} \cos \theta$$

$$\text{or, } 16 = 6\sqrt{21} \cos \theta$$

$$\text{or, } \cos \theta = \frac{16}{6\sqrt{21}}$$

$$\text{or, } \theta = \cos^{-1}\left(\frac{16}{6\sqrt{21}}\right)$$

$$\therefore \theta = 54.415^\circ$$

(Ans.)

Divergence →

(15) If $\vec{A} = x^2z\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}$, find $\vec{\nabla} \cdot \vec{A}$ (or $\text{div } \vec{A}$) at the point $(1, -1, 1)$

Solution: Given that,

$$\vec{A} = x^2z\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}$$

Also we know that,

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

$$\begin{aligned}\therefore \vec{\nabla} \cdot \vec{A} &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (x^2z\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}) \\ &= \frac{\partial}{\partial x}(x^2z) + \frac{\partial}{\partial y}(-2y^3z^2) + \frac{\partial}{\partial z}(xy^2z) \\ &= 2xz - 6y^2z^2 + xy^2\end{aligned}$$

At the point $(1, -1, 1)$,

$$\begin{aligned}\vec{\nabla} \cdot \vec{A} &= 2 \cdot 1 \cdot 1 - 6(-1)^2 \cdot 1^2 + 1 \cdot (-1)^2 \\ &= 2 - 6 + 1 \\ &= -3 \quad (\text{Ans.})\end{aligned}$$

(16) Given $\phi = 2x^3y^2z^4$ (a) Find $\vec{\nabla} \cdot \vec{\nabla} \phi$ (or $\text{div grad } \phi$)

(b) Show that $\vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi$, where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ denotes the Laplacian operator.

Solution: Given that,

$$\phi = 2x^3y^2z^4$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Also know that,

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

(a) Now,

$$\begin{aligned}\vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (2x^3 y^2 z^4) \\&= \hat{i} \frac{\partial}{\partial x} (2x^3 y^2 z^4) + \hat{j} \frac{\partial}{\partial y} (2x^3 y^2 z^4) + \hat{k} \frac{\partial}{\partial z} (2x^3 y^2 z^4) \\&= 6x^2 y^2 z^4 \hat{i} + 4x^3 y z^4 \hat{j} + 8x^3 y^2 z^3 \hat{k}\end{aligned}$$

Also,

$$\begin{aligned}\vec{\nabla} \cdot \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (6x^2 y^2 z^4 \hat{i} + 4x^3 y z^4 \hat{j} + 8x^3 y^2 z^3 \hat{k}) \\&= \frac{\partial}{\partial x} (6x^2 y^2 z^4) + \frac{\partial}{\partial y} (4x^3 y z^4) + \frac{\partial}{\partial z} (8x^3 y^2 z^3) \\&= 12x y^2 z^4 + 4x^3 z^4 + 24x^3 y^2 z^2\end{aligned}$$

(Ans.)

$$\begin{aligned}\textcircled{b} \text{ Now, } \vec{\nabla} \cdot \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \left\{ \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \right\} \\&= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \\&= \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} \right) \\&= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \\&= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi \\&= \nabla^2 \phi\end{aligned}$$

Hence, $\vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi$

[shown]

①7 Prove that, $\vec{\nabla}^2 \left(\frac{1}{r} \right) = 0$

Solution: We know that,

$$\vec{\nabla}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

Now,

$$\begin{aligned} \vec{\nabla}^2 \left(\frac{1}{r} \right) &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) \\ &= \frac{\partial^2}{\partial x^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) + \frac{\partial^2}{\partial y^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) + \frac{\partial^2}{\partial z^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) \quad \text{--- ①} \end{aligned}$$

Again,

$$\begin{aligned} \frac{\partial^2}{\partial x^2} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) &= \frac{\partial}{\partial x} \left\{ \frac{\partial}{\partial x} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) \right\} \\ &= \frac{\partial}{\partial x} \left\{ \frac{-1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2x \right\} \\ &= \frac{\partial}{\partial x} \left\{ -x \cdot (x^2 + y^2 + z^2)^{-3/2} \right\} \\ &= -x \left(\frac{-3}{2} \right) (x^2 + y^2 + z^2)^{-5/2} \cdot 2x - (x^2 + y^2 + z^2)^{-3/2} \\ &= 3x^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\ &= \frac{3x^2}{(x^2 + y^2 + z^2)^{5/2}} - \frac{1}{(x^2 + y^2 + z^2)^{3/2}} \\ &= \frac{3x^2 - x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} \\ &= \frac{2x^2 - y^2 - z^2}{(x^2 + y^2 + z^2)^{5/2}} \end{aligned}$$

Similarly $\frac{\partial^2}{\partial y^2} \left(\frac{1}{\sqrt{x^2+y^2+z^2}} \right) = \frac{2y^2 - z^2 - x^2}{(x^2+y^2+z^2)^{5/2}}$

$$\frac{\partial^2}{\partial z^2} \left(\frac{1}{\sqrt{x^2+y^2+z^2}} \right) = \frac{2z^2 - y^2 - x^2}{(x^2+y^2+z^2)^{5/2}}$$

From equation (1) we get,

$$\vec{\nabla}^2 \left(\frac{1}{r} \right) = \frac{2x^2 - y^2 - z^2}{(x^2+y^2+z^2)^{5/2}} + \frac{2y^2 - z^2 - x^2}{(x^2+y^2+z^2)^{5/2}} + \frac{2z^2 - y^2 - x^2}{(x^2+y^2+z^2)^{5/2}}$$

$$= \frac{2x^2 - y^2 + z^2 + 2y^2 - z^2 - x^2 + 2z^2 - y^2 - x^2}{(x^2+y^2+z^2)^{5/2}}$$

$$= \frac{0}{(x^2+y^2+z^2)^{5/2}}$$

$$= 0$$

Hence, $\vec{\nabla}^2 \left(\frac{1}{r} \right) = 0$

(10) $\vec{\nabla} \cdot \vec{\nabla} \left(\frac{1}{r} \right) = 0$

19) Prove $\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right) = 0$

Solution:

$$\begin{aligned}\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right) &= \vec{\nabla} \cdot (r^{-3} \vec{r}) \\ &= \vec{\nabla} \cdot (\phi \vec{A}) \quad [\text{Let, } r^{-3} = \phi, \vec{r} = \vec{A}] \\ &= (\vec{\nabla} \phi) \cdot \vec{A} + \phi (\vec{\nabla} \cdot \vec{A}) \quad [\because \vec{\nabla} \cdot (\phi \vec{A}) = (\vec{\nabla} \phi) \cdot \vec{A} + \phi (\vec{\nabla} \cdot \vec{A})] \\ &= (\vec{\nabla} r^{-3}) \cdot \vec{r} + r^{-3} (\vec{\nabla} \cdot \vec{r}) \\ &= -3r^{-5} \vec{r} \cdot \vec{r} + r^{-3} \times 3 \quad [\because \vec{\nabla} r^n = n r^{n-2} \vec{r}] \\ &= -3r^{-5} r^2 + 3r^{-3} \quad \vec{\nabla} \cdot \vec{r} = 3 \\ &= -3r^{-3} + 3r^{-3} \\ &= 0\end{aligned}$$

$$\therefore \vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right) = 0 \quad [\text{Proved}]$$

22) Prove that the vector $\vec{V} =$

② Determine the constant a so that the vector $\vec{V} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal.

Solution: We know that,

A vector $\vec{V}$ is solenoidal if its divergence is

zero; $\vec{\nabla} \cdot \vec{V} = 0$

$$\text{or, } \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k} = 0$$

$$\text{or, } \frac{\partial}{\partial x} (x+3y) + \frac{\partial}{\partial y} (y-2z) + \frac{\partial}{\partial z} (x+az) = 0$$

$$\text{or, } 1 + 1 + a = 0$$

$$\text{or, } 2 + a = 0$$

$$\therefore a = -2$$

(Ans.)

The Curl →

(23) If $\vec{A} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$, find

$\vec{\nabla} \times \vec{A}$ (or curl $\vec{A}$) at the point $(1, -1, 1)$.

Solution: Given that,

$$\vec{A} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$$

Also we know that,

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

Now,

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(2yz^4) - \frac{\partial}{\partial z}(-2x^2yz) \right] - \hat{j} \left[\frac{\partial}{\partial x}(2yz^4) - \frac{\partial}{\partial z}(xz^3) \right] + \hat{k} \left[\frac{\partial}{\partial x}(-2x^2yz) - \frac{\partial}{\partial y}(xz^3) \right]$$

$$= \hat{i} [2z^4 + 2x^2y] - \hat{j} [-3xz^2] + \hat{k} [-4xyz]$$

$$= (2z^4 + 2x^2y)\hat{i} + 3xz^2\hat{j} - 4xyz\hat{k}$$

At the point $(1, -1, 1)$,

$$\vec{\nabla} \times \vec{A} = [2 \cdot 1 + 2 \cdot (1)^2 \cdot (-1)]\hat{i} + [3(1) \cdot (1)^2]\hat{j} - [4 \cdot 1 \cdot (-1) \cdot 1]\hat{k}$$

$$\therefore \vec{\nabla} \times \vec{A} = 3\hat{j} + 4\hat{k} \text{ (Ans.)}$$

24) If $\vec{A} = x^2y\hat{i} - 2xz\hat{j} + 2yz\hat{k}$, find curl curl $\vec{A}$.

Solution: Given that,

$$\vec{A} = x^2y\hat{i} - 2xz\hat{j} + 2yz\hat{k}$$

Also we know that,

$$\text{Curl curl } \vec{A} = \nabla \times (\nabla \times \vec{A})$$

$$\text{Now, } \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & -2xz & 2yz \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (2yz) - \frac{\partial}{\partial z} (-2xz) \right] - \hat{j} \left[\frac{\partial}{\partial x} (2yz) - \frac{\partial}{\partial z} (x^2y) \right] + \hat{k} \left[\frac{\partial}{\partial x} (-2xz) - \frac{\partial}{\partial y} (x^2y) \right]$$

$$= \hat{i} [2z + 2x] - \hat{j} [0] + \hat{k} [-2z - x^2]$$

$$\therefore \nabla \times \vec{A} = (2x + 2z)\hat{i} - (x^2 + 2z)\hat{k}$$

$$\text{Again, } \nabla \times (\nabla \times \vec{A}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (2x+2z) & 0 & -(x^2+2z) \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (-x^2 - 2z) \right] - \hat{j} \left[\frac{\partial}{\partial x} (-x^2 - 2z) - \frac{\partial}{\partial z} (2x + 2z) \right] + \hat{k} \left[0 - \frac{\partial}{\partial y} (2x + 2z) \right]$$

$$= 0\hat{i} - \hat{j} [-2x - 2] + 0\hat{k}$$

$$= (2x + 2)\hat{j} \quad (\text{Ans.})$$

32 Find constants a, b, c so that $\vec{V} = (x+2y+az)\hat{i} + (bx-3y-z)\hat{j} + (4x+cy+2z)\hat{k}$ is irrotational.

Solution: Given that,

$$\vec{V} = (x+2y+az)\hat{i} + (bx-3y-z)\hat{j} + (4x+cy+2z)\hat{k}$$

We know that, $\nabla \times \vec{V} = 0$ if $\vec{V}$ is irrotational

$\vec{V}$ is called irrotational if $\nabla \times \vec{V} = 0$

Now,

$$\begin{aligned} \nabla \times \vec{V} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x+2y+az) & (bx-3y-z) & (4x+cy+2z) \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y} (4x+cy+2z) - \frac{\partial}{\partial z} (bx-3y-z) \right] - \hat{j} \left[\frac{\partial}{\partial x} (4x+cy+2z) - \frac{\partial}{\partial z} (x+2y+az) \right] + \hat{k} \left[\frac{\partial}{\partial x} (bx-3y-z) - \frac{\partial}{\partial y} (x+2y+az) \right] \\ &= \hat{i} [c+1] - \hat{j} [4-a] + \hat{k} [b-2] \end{aligned}$$

This equals zero (0) when, $c+1=0$; $4-a=0$; $b-2=0$
or, $c=-1$ or, $a=4$; $b=2$

$\therefore a=4, b=2, c=-1$. (Ans.)

$$\therefore \vec{V} = (x+2y+4z)\hat{i} + (2x-3y-z)\hat{j} + (4x-y+2z)\hat{k}$$

(Ans.)

Line Integral.

⑥ If $\vec{A} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$, evaluate $\int_C \vec{A} \cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the following paths c:

a. $x=t, y=t^2, z=t^3$

b. The straight lines from $(0,0,0)$ to $(1,0,0)$, then to $(1,1,0)$; and then to $(1,1,1)$

c. The straight line joining $(0,0,0)$ and $(1,1,1)$.

Solution: Given that,

$$\vec{A} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$$

We know that,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\begin{aligned}\therefore \int_C \vec{A} \cdot d\vec{r} &= \int_C \{ (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k} \} (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int_C (3x^2 + 6y)dx - 14yz dy + 20xz^2 dz\end{aligned}$$

⑦ Given that, $x=t, y=t^2, z=t^3$ points $(0,0,0)$ and $(1,1,1)$ correspond to $t=0$ and $t=1$ respectively.

$$\begin{aligned}\therefore \vec{A} &= (3t^2 + 6t^2)\hat{i} - 14t^2 t^3 \hat{j} + 20t \cdot (t^3)^2 \hat{k} \\ &= 9t^2 \hat{i} - 14t^5 \hat{j} + 20t^7 \hat{k}\end{aligned}$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$= t\hat{i} + t^2\hat{j} + t^3\hat{k}$$

$$\therefore d\vec{r} = (\hat{i} + 2t\hat{j} + 3t^2\hat{k}) dt$$

$$\begin{aligned}
 \therefore \int_C \vec{A} \cdot d\vec{r} &= \int_0^1 (9t^2 \hat{i} - 14t^5 \hat{j} + 20t^7 \hat{k}) \cdot (\hat{i} + 2t \hat{j} + 3t^2 \hat{k}) dt \\
 &= \int_0^1 (9t^2 - 28t^6 + 60t^9) dt \\
 &= \left[9 \frac{t^3}{3} - 28 \frac{t^7}{7} + 60 \frac{t^{10}}{10} \right]_0^1 \\
 &= [3(1)^3 - 4(1)^7 + 6(1)^{10} - 0 + 0 - 0] \\
 &= 5.
 \end{aligned}$$

(Ans)

B Along the straight line from $(0,0,0)$ to $(1,0,0)$ $y=0, z=0$; $dy=0, dz=0$ while x varies from 0 to 1

Then the integral over path is $\rightarrow$

$$\begin{aligned}
 &\int_{x=0}^1 (3x^2 + 6 \cdot 0) dx - 14 \cdot 0 \cdot 0 \cdot 0 + 20x \cdot 0 \cdot 0 \\
 &= \int_{x=0}^1 3x^2 dx \\
 &= \left[3 \cdot \frac{x^3}{3} \right]_{x=0}^1 \\
 &= 1.
 \end{aligned}$$

(Ans).

Along the straight line from $(1,0,0)$ to $(1,1,0)$

$x=1, z=0$; $dx=0, dz=0$ while y varies from 0 to 1

$$\begin{aligned}
 \therefore \int_{y=0}^1 (3 \cdot 1^2 + 6y) \cdot 0 - 14y(0) dy + 20 \cdot 1 \cdot 0^2 \cdot 0 \\
 = 0
 \end{aligned}$$

Along the straight line from $(1, 1, 0)$ to $(1, 1, 1)$
 $x=1, y=1, dx=0, dy=0$ while z varies from 0 to 1.

Then the integral over this part of path is

$$\begin{aligned} & \int_{z=0}^1 \{ 3(1)^2 + 6(1) \} 0 - 14 \cdot 1 \cdot z \cdot 0 + 20 \cdot 1 \cdot z^2 dz \\ &= \int_{z=0}^1 20z^2 dz \\ &= \left[20 \cdot \frac{z^3}{3} \right]_{z=0}^1 \\ &= \frac{20}{3} \text{ (Ans.)} \end{aligned}$$

(c) The straight line joining $(0, 0, 0)$ and $(1, 1, 1)$ is given parametric form by $x=t, y=t, z=t$

Then,

$$\begin{aligned} \int_C \vec{A} \cdot d\vec{r} &= \int_{t=0}^1 (3t^2 + 6t) \cdot dt - 14(t)(t) dt + 20t \cdot t^2 dt \\ &= \int_0^1 (3t^2 + 6t) dt - 14t^2 dt + 20t^3 dt \\ &= \int_0^1 (20t^3 - 11t^2 + 6t) dt \\ &= \left[20 \cdot \frac{t^4}{4} - 11 \cdot \frac{t^3}{3} + 6 \cdot \frac{t^2}{2} \right]_0^1 \\ &= 5 - \frac{11}{3} + 3 \\ &= \frac{13}{3} \text{ (Ans.)} \end{aligned}$$

— 0 —

⑦ Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = (t^2 + 1)$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$

Solution: Given that,

$$\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$$

$$\text{also } x = t^2 + 1, y = 2t^2, z = t^3.$$

$$\therefore dx = 2t dt, dy = 4t dt \text{ and } dz = 3t^2 dt$$

Again we know that,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

Now, the total work is done by,

$$W = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_C (3xy\hat{i} - 5z\hat{j} + 10x\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \int_C (3xy dx - 5z dy + 10x dz)$$

$$= \int_1^2 \{ 3(t^2 + 1)(2t^2)(2t) dt - 5t^3(4t) dt + 10(t^2 + 1)3t^2 \cdot dt \}$$

$$= \int_1^2 \{ (3t^2 + 3) \cdot 4t^3 dt - 20t^4 dt + 30t^2(t^2 + 1) dt \}$$

$$= \int_1^2 (12t^5 + 12t^3 - 20t^4 + 30t^4 + 30t^2) dt$$

$$= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2) dt$$

$$\begin{aligned}
 &= \left[\frac{12 \cdot t^6}{6} + 10 \cdot \frac{t^5}{5} + 12 \cdot \frac{t^4}{4} + 30 \cdot \frac{t^3}{3} \right]_1^2 \\
 &= \{ 2 \cdot (2)^6 + 2 \cdot (2)^5 + 3(2)^4 + 10(2)^3 \} - \{ 2 + 2 + 3 + 10 \} \\
 &= 320 - 17 \\
 &= 303 \quad (\text{Ans.})
 \end{aligned}$$

⑧ If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve in the xy plane, $y = 2x^2$ from $(0,0)$ to $(1,2)$.

Solution: Given that,

$$\vec{F} = 3xy\hat{i} - y^2\hat{j}$$

$$\text{also } y = 2x^2$$

$$\therefore dy = 4x dx$$

$$\text{Again we know that, } \vec{r} = x\hat{i} + y\hat{j} \\ = dx\hat{i} + dy\hat{j}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_C (3xy\hat{i} - y^2\hat{j}) (dx\hat{i} + dy\hat{j})$$

$$= \int_C (3xy dx - y^2 dy)$$

Now substituting $y = 2x^2$ and points $(0,0)$ and $(1,2)$ correspond to $t = 0$ and $t = 1$ respectively. Then x goes from 0 to 1.

$$\int_C \vec{F} \cdot d\vec{r} = \int_{x=0}^1 3x(2x^2) dx - (2x^2)^2 4x dx$$

$$= \int_0^1 6x^3 dx - 16x^5 dx$$

$$\begin{aligned}
 &= \int_0^1 (6x^3 - 16x^5) dx \\
 &= \left[6 \frac{x^4}{4} - 16 \frac{x^6}{6} \right]_0^1 \\
 &= \left[\frac{6}{4} (1)^4 - \frac{16}{6} (1)^6 \right] \\
 &= \frac{-7}{6} \quad (\text{Ans.})
 \end{aligned}$$

⑨ Find the work done in moving a particle once around a circle C in the xy plane, if the circle has center at the origin and radius 3 and if the force field is given by $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$

Solution: Given that,

$$\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$$

Also we know that,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

According to Question, $z = 0$. So that

$$\vec{F} = (2x - y)\hat{i} + (x + y)\hat{j} + (3x - 2y)\hat{k}$$

$$\text{and } \vec{r} = x\hat{i} + y\hat{j}$$

$$\therefore d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\text{Now, the work done} = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_C \{(2x - y)\hat{i} + (x + y)\hat{j} + (3x - 2y)\hat{k}\} \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C \{ (2x-y)dx + (x+y)dy \}$$

Again given that the circle has centre at the origin $(0,0)$ and radius 3

Let the parametric eqn of the circle arc,
 $x = 3\cos t$ and $y = 3\sin t$, where $t=0$ to $t=2\pi$.

$$\therefore dx = -3\sin t dt \text{ and } dy = 3\cos t dt$$

Therefore,

$$\text{work done} = \int_0^{2\pi} \{ (6\cos t - 3\sin t)(-3\sin t dt) + (3\cos t + 3\sin t) 3\cos t dt \}$$

$$= \int_0^{2\pi} \{ -18\cos t \sin t + 9\sin^2 t + 9\cos^2 t + 9\sin t \cdot \cos t \} dt$$

$$= \int_0^{2\pi} \{ 9(\sin^2 t + \cos^2 t) - 9\sin t \cdot \cos t \} dt$$

$$= \int_0^{2\pi} \{ 9 - \frac{9}{2} \cdot 2\sin t \cos t \} dt$$

$$= \int_0^{2\pi} (9 - \frac{9}{2} \sin 2t) dt$$

$$= \left[9t - \frac{9}{2} \cdot \frac{\cos 2t}{2} \right]_0^{2\pi}$$

$$= (9 \cdot 2\pi - \frac{9}{4} \cos 4\pi) - (0 - \frac{9}{4})$$

$$= 18\pi - \frac{9}{4} + \frac{9}{4}$$

$$= 18\pi$$

(Ans).

16) If $\phi = 2xyz^2$; $\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and C is the curve $x = t^2$; $y = 2t$, $z = t^3$ from $t=0$ to $t=1$ evaluate the line integrals.

a) $\int_C \phi d\vec{r}$ b) $\int_C \vec{F} \times d\vec{r}$

Solution: Given that,

$$\phi = 2xyz^2$$

$$\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$$

$$\text{Also, } x = t^2; y = 2t; z = t^3$$

$$\therefore dx = 2t dt, dy = 2 dt, dz = 3t^2 dt$$

$$\text{Also we know that, } \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$= (2t\hat{i} + 2\hat{j} + 3t^2\hat{k}) dt$$

$$\begin{aligned} \text{a) } \phi &= 2xyz^2 \\ &= 2t^2(2t)(t^3)^2 \\ &= 4t^9 \end{aligned}$$

$$\begin{aligned} \text{Now, } \int_C \phi d\vec{r} &= \int_{t=0}^1 4t^9 (2t\hat{i} + 2\hat{j} + 3t^2\hat{k}) dt \\ &= \int_0^1 (8t^{10}\hat{i} + 8t^9\hat{j} + 12t^{11}\hat{k}) dt \\ &= \left[8 \frac{t^{11}}{11}\hat{i} + 8 \frac{t^{10}}{10}\hat{j} + 12 \frac{t^{12}}{12}\hat{k} \right]_0^1 \\ &= \frac{8}{11}\hat{i} + \frac{8}{10}\hat{j} + \hat{k} \end{aligned}$$

(Ans.)

$$\begin{aligned}\textcircled{b} \vec{F} &= xy \hat{i} - z \hat{j} + x^2 \hat{k} \\ &= (t^2 \cdot 2t) \hat{i} - t^3 \hat{j} + t^4 \hat{k} \\ &= 2t^3 \hat{i} - t^3 \hat{j} + t^4 \hat{k}\end{aligned}$$

$$\begin{aligned}\text{Now, } \vec{F} \times d\vec{r} &= (2t^3 \hat{i} - t^3 \hat{j} + t^4 \hat{k}) \times (2t \hat{i} + 2 \hat{j} + 3t^2 \hat{k}) dt \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2t^3 & -t^3 & t^4 \\ 2t dt & 2 dt & 3t^2 dt \end{vmatrix} \\ &= \hat{i} \{ -t^3(3t^2 dt) - t^4(2 dt) \} - \hat{j} \{ 2t^3(3t^2 dt) - t^4(2t dt) \} + \hat{k} \{ 2t^3(2 dt) - (-t^3)(2t dt) \}\end{aligned}$$

$$\begin{aligned}&= \hat{i}(-3t^5 dt - 2t^4 dt) - \hat{j}(6t^5 dt - 2t^5 dt) + \hat{k}(4t^3 dt + 2t^4 dt) \\ &= \hat{i}(-3t^5 dt - 2t^4 dt) - 4t^5 dt \hat{j} + (4t^3 dt + 2t^4 dt) \hat{k}\end{aligned}$$

$$\therefore \int_C \vec{F} \times d\vec{r} = \int_{t=0}^1 \{ (-3t^5 - 2t^4) \hat{i} - 4t^5 \hat{j} + (4t^3 dt + 2t^4) \hat{k} \} dt$$

$$= \left[\left(-3 \frac{t^6}{6} - 2 \frac{t^5}{5} \right) \hat{i} - \left(4 \frac{t^6}{6} \hat{j} \right) + \left(4 \frac{t^4}{4} + 2 \frac{t^5}{5} \right) \hat{k} \right]_0^1$$

$$= \left[\left(-\frac{1}{2} - \frac{2}{5} \right) \hat{i} - \frac{2}{3} \hat{j} + \left(1 + \frac{2}{5} \right) \hat{k} - 0 \right]$$

$$= \left(-\frac{1}{2} - \frac{2}{5} \right) \hat{i} - \frac{2}{3} \hat{j} + \left(1 + \frac{2}{5} \right) \hat{k}$$

$$= -\frac{9}{10} \hat{i} - \frac{2}{3} \hat{j} + \frac{7}{5} \hat{k}$$

(Ans.)

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*From PDF →

Ques: (66) How much work is accomplished by the force $\vec{F}(x, y) = xy\hat{i} + y\hat{j}$ in pushing a particle from $(0, 0)$ to $(3, 9)$ along the parabola $y = x^2$?

Solution: Given that,

$$\vec{F}(x, y) = xy\hat{i} + y\hat{j}$$

Also know that, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Along the curve, $y = x^2$

$$\vec{F} = x \cdot x^2 \hat{i} + x^2 \hat{j} \Rightarrow x^3 \hat{i} + x^2 \hat{j}$$

$$\text{and } \vec{r} = x\hat{i} + x^2\hat{j} \quad [\text{As } z=0]$$

$$\therefore d\vec{r} = dx\hat{i} + 2x dx \hat{j}$$

Now the work done along the curve $y = x^2$ from $(0, 0)$ to $(3, 9)$.

$$W = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_0^3 (x^3 \hat{i} + x^2 \hat{j}) \cdot (dx\hat{i} + 2x dx \hat{j})$$

$$= \int_0^3 (x^3 + 2x^3) dx \Rightarrow \int_0^3 (3x^3) dx$$

$$= 3 \left[\frac{x^4}{4} \right]_0^3$$

$$= \frac{3}{4} \times (3)^4 - 0$$

$$= \frac{243}{4} \quad (\text{Ans.})$$

Green's Theorem:

Let R be a simple connected region with a piece wise smooth boundary C , oriented counter clockwise. If $M(x,y)$ and $N(x,y)$ be continuous and have continuous partial derivatives in an open region containing R , then

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Ques: Verify Green's theorem in the plane for $\oint (xy + x^2) dx + xy^2 dy$ where C is the closed curve of the region bounded by $y=x$ and $y^2=x$.

Solution: Given that,

$$y = x \quad \text{--- (1)}$$

$$y^2 = x \quad \text{--- (2)}$$

Now from (1) and (2) we get,

$$x^2 = x$$

$$\text{or, } x^2 - x = 0$$

$$\text{or, } x(x-1) = 0$$

$$\therefore x = 0, 1$$

$$\text{So that, } y = 0, 1$$

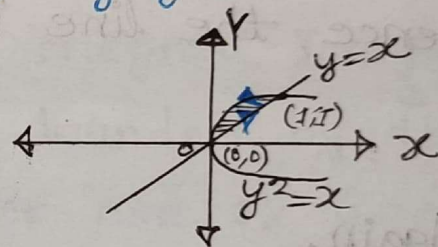
Therefore, the intersect point are $(0,0)$ and $(1,1)$

Now, along the line $y=x$, the line integral

$$I_1 = \int_0^1 [(x \cdot x + x^2) dx + x \cdot x^2 dx] \quad \left[\begin{array}{l} \text{As } y = x \\ \text{or, } dy = dx \end{array} \right]$$

$$= \int_0^1 (2x^2 + x^3) dx$$

$$= \frac{2}{3} + \frac{1}{4} \Rightarrow \frac{11}{12}$$



Again along the curve $y^2 = x$ or, $y = \sqrt{x}$ the line integral,

$$I_2 = \int_1^0 \left\{ (x \cdot \sqrt{x} + x^2) dx + x \cdot x \cdot \frac{1}{2\sqrt{x}} dx \right\} \left[\text{As } y = \sqrt{x} \right. \\ \left. \text{or, } dy = \frac{1}{2\sqrt{x}} dx \right]$$

$$= \int_1^0 \left(x^{3/2} + x^2 + \frac{x^{3/2}}{2} \right) dx$$

$$= \left[\frac{x^{5/2}}{5/2} + \frac{x^3}{3} + \frac{1}{2} \cdot \frac{x^{5/2}}{5/2} \right]_1^0$$

$$= - \left(\frac{2}{5} + \frac{1}{3} + \frac{1}{5} \right)$$

$$= - \frac{14}{15}$$

Hence, the line integral is $I = I_1 + I_2$

$$= \frac{11}{12} - \frac{14}{15}$$

$$= \frac{-1}{60}$$

Again,

according to Ques, $M = xy + x^2$ and $N = xy^2$

$$\therefore \frac{\partial M}{\partial y} = x \text{ and } \frac{\partial N}{\partial x} = y^2$$

Now,

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

$$= \iint_R (y^2 - x) dA$$

$$= \int_0^1 \int_{y=x}^{\sqrt{x}} (y^2 - x) dy dx$$

$$= \int_0^1 \left[\frac{y^3}{3} - xy \right]_x^{\sqrt{x}} dx$$

$$= \int_0^1 \left(\frac{1}{3} x^{3/2} - x^{3/2} - \frac{1}{3} x^3 + x^2 \right) dx$$

$$= \left[\frac{1}{3} \cdot \frac{x^{5/2}}{5/2} - \frac{x^{5/2}}{2/5} - \frac{1}{3} \cdot \frac{x^4}{4} + \frac{1}{3} x^3 \right]_0^1$$

$$= \frac{2}{15} - \frac{2}{5} - \frac{1}{12} + \frac{1}{3}$$

$$= \frac{-1}{60}$$

Therefore, for the given question,

$$\oint_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \frac{-1}{60}$$

Hence, Green's theorem verified.

(Ans.)

From PDF →

Example : 87 Evaluate the line integral $I = \oint_C (2x - y)dy$ round the region bounded by the curve $y = x^2$ and $x = y^2$ by the use of Green's theorem.

Solution: Given that,

$$y = x^2 \quad \text{--- (1)}$$

$$x = y^2 \quad \text{--- (2)}$$

Now from (1) and (2) we get,

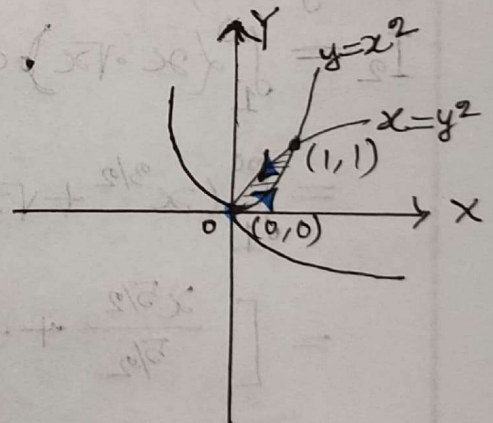
$$x = (x^2)^2$$

$$\text{or, } x^4 - x = 0$$

$$\text{or, } x(x^3 - 1) = 0$$

$$\therefore x = 0, 1$$

$$\text{So that, } y = 0, 1$$



Therefore, the intersect point are $(0,0)$ and $(1,1)$.

Now, along the curve $y = x^2$ the line integral

$$\begin{aligned} I_1 &= \int_0^1 [(x \cdot x^2) dx + (2x - x^2) 2x dx] \quad [\text{As } y = x^2 \\ &\quad \text{or, } dy = 2x dx] \\ &= \int_0^1 [(x^3 + 4x^2 - 2x^3) dx] \\ &= \int_0^1 [(-x^3 + 4x^2) dx] \\ &= \left[\frac{-x^4}{4} + 4 \cdot \frac{x^3}{3} \right]_0^1 \end{aligned}$$

$$= \left[-\frac{1}{4} + \frac{4}{3} \right] = \frac{13}{12}$$

Again, along the curve $x = y^2$ or, $\sqrt{x} = y$ the line integral,

$$\begin{aligned} I_2 &= \int_1^0 \left\{ (x \cdot \sqrt{x}) dx + (2x - \sqrt{x}) \frac{1}{2\sqrt{x}} dx \right\} \quad [\text{As } y = \sqrt{x} \\ &\quad \text{or, } dy = \frac{1}{2\sqrt{x}} dx] \\ &= \int_1^0 \left(x^{3/2} + \sqrt{x} - \frac{1}{2} \right) dx \quad (1) \\ &= \left[\frac{x^{5/2}}{5/2} + \frac{x^{3/2}}{3/2} - \frac{x}{2} \right]_1^0 \\ &= - \left[\frac{2}{5} + \frac{2}{3} - \frac{1}{2} \right] \\ &= - \frac{17}{30} \end{aligned}$$

Hence, the line integral is $I = I_1 + I_2$

$$\begin{aligned} &= \frac{13}{12} - \frac{17}{30} \\ &= \frac{31}{60} \end{aligned}$$

Again, according to ques, $M=xy$ and $N=(2x-y)$
 $\therefore \frac{\partial M}{\partial y} = x$ and $\frac{\partial N}{\partial x} = 2$

Now, $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$
 $= \iint_R (2-x) dA$
 $= \int_0^1 \int_{y=x^2}^{\sqrt{x}} (2-x) dy dx$
 $= \int_0^1 [2y - xy]_{x^2}^{\sqrt{x}} dx$
 $= \int_0^1 [2\sqrt{x} - x\sqrt{x} - 2x^2 + x^3] dx$
 $= \int_0^1 [2\sqrt{x} - x^{3/2} - 2x^2 + x^3] dx$
 $= \left[2 \frac{x^{3/2}}{3/2} - \frac{x^{5/2}}{5/2} - \frac{2}{3} x^3 + \frac{x^4}{4} \right]_0^1$
 $= \frac{4}{3} - \frac{2}{5} - \frac{2}{3} + \frac{1}{4}$
 $= \frac{31}{60}$

Therefore, for the given question,

$$\oint_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \frac{31}{60}$$

Hence, Green's theorem verified.
(Ans.)

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Volume Integral

Ques: State Divergence theorem (Gauss Theorem)

Solution:

statement: For a close surface, enclosing a region V in a vector field $\vec{F}$, then $\int_V \text{div } \vec{F} dv = \int_S \vec{F} \cdot d\vec{S}$ which is called the Divergence theorem.

Q.77: Verify the Divergence theorem for the vector field $\vec{F} = x^2\hat{i} + z\hat{j} + y\hat{k}$ taken over the region bounded by the planes $z=0, z=2, x=0, x=1, y=0, y=3$

Solution: Given that,

$$\vec{F} = x^2\hat{i} + z\hat{j} + y\hat{k}$$

Now,

$$\text{div } \vec{F} = \nabla \cdot \vec{F}$$

$$= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^2\hat{i} + z\hat{j} + y\hat{k})$$

$$= \frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial y} (z) + \frac{\partial}{\partial z} (y)$$

$$\therefore \text{div } \vec{F} = 2x$$

$$\therefore \int_V \operatorname{div} \vec{F} dv = \int_{x=0}^1 \int_{y=0}^3 \int_{z=0}^2 2x dz dy dx$$

$$= \int_{x=0}^1 \int_{y=0}^3 [2xz]_{z=0}^2 dy dx$$

$$= \int_{x=0}^1 \int_{y=0}^3 4x dy dx$$

$$= \int_{x=0}^1 [4xy]_{y=0}^3 dx$$

$$= \int_{x=0}^1 12x dx$$

$$= \left[\frac{12x^2}{2} \right]_{x=0}^1$$

$$\therefore \int_V \operatorname{div} \vec{F} dv = 6$$

Hence, $\int_V \operatorname{div} \vec{F} dv = 6$

Again we know that,

$$\int_S \vec{F} \cdot d\vec{s} = \int_S \vec{F} \cdot \hat{n} ds$$

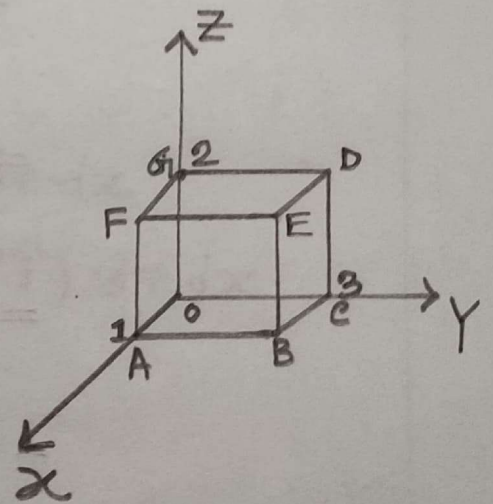
Now,

S_1 (OABC):

Where $z=0$, $\hat{n} = -\hat{k}$ and $ds_1 = dx dy$

$$\therefore \int_{S_1} \vec{F} \cdot \hat{n} \cdot ds_1 = \int_{S_1} (x^2 \hat{i} + 0 \hat{j} + y \hat{k}) (-\hat{k}) \cdot ds_1$$

$$= \int_{y=0}^3 \int_{x=0}^1 (-y) dx dy$$



$$= \int_{y=0}^3 [-yx]_{x=0}^1 dy$$

$$= \int_{y=0}^3 (-y) dy$$

$$= \left[-\frac{y^2}{2} \right]_{y=0}^3$$

$$= -\frac{9}{2}$$

Also, $S_2(ED \cap F)$:

where, $z=2$, $\hat{n} = \hat{k}$ and $ds_2 = dxdy$

$$\therefore \int_{S_2} \vec{F} \cdot \hat{n} ds_2 = \int_{S_2} (x^2 \hat{i} + 2\hat{j} + y\hat{k}) \cdot (\hat{k}) ds_2$$

$$= \int_{y=0}^3 \int_{x=0}^1 y dxdy$$

$$= \int_{y=0}^3 [xy]_{x=0}^1 dy$$

$$= \int_{y=0}^3 y dy$$

$$= \left[\frac{y^2}{2} \right]_{y=0}^3$$

$$= \frac{9}{2}$$

Also, $S_3(OAFG)$:

where, $y=0$; $\hat{n} = -\hat{j}$ and $ds_3 = dz dx$

$$\therefore \int_{S_3} \vec{F} \cdot \hat{n} \cdot dS_3 = \int_{S_3} (x^2 \hat{i} + z \hat{j} + 0 \hat{k}) \cdot (-\hat{j}) \cdot dS_3$$

$$= \int_{x=0}^1 \int_{z=0}^2 (-z) \cdot dz \cdot dx$$

$$= \int_{x=0}^1 \left[-\frac{z^2}{2} \right]_{z=0}^2 dx$$

$$= \int_{x=0}^1 (-2) dx$$

$$= \left[-2x \right]_{x=0}^1$$

$$= -2$$

Also, $S_4(BCDE)$:

where, $y=3$; $\hat{n} = \hat{j}$ and $ds_4 = dz dx$

$$\therefore \int_{S_4} \vec{F} \cdot \hat{n} \cdot dS_4 = \int_{S_4} (x^2 \hat{i} + z \hat{j} + 3 \hat{k}) \cdot (\hat{j}) \cdot dz dx$$

$$= \int_{x=0}^1 \int_{z=0}^2 z \cdot dz \cdot dx$$

$$= \int_{x=0}^1 \left[\frac{z^2}{2} \right]_{z=0}^2 dx$$

$$= \int_{x=0}^1 2 dx$$

$$= \left[2x \right]_{x=0}^1$$

$$= 2$$

Also, $S_5(ABEF)$:

where $x=1$, $\hat{n}=\hat{i}$ and $ds_5 = dy dz$

$$\therefore \int_{S_5} \vec{F} \cdot \hat{n} ds_5 = \int_{S_5} (\hat{i} + z\hat{j} + y\hat{k}) \cdot (\hat{i}) dy dz$$

$$= \int_{z=0}^2 \int_{y=0}^3 1 dy dz$$

$$= \int_{z=0}^2 [y]_{y=0}^3 dz$$

$$= \int_{z=0}^2 3 dz$$

$$= [3z]_{z=0}^2$$

$$= 6$$

Also, $S_6(OC DG)$:

where $x=0$, $\hat{n}=-\hat{i}$ and $ds_6 = dy dz$

$$\therefore \int_{S_6} \vec{F} \cdot \hat{n} ds_6 = \int_{S_6} (0\hat{i} + z\hat{j} + y\hat{k}) \cdot (-\hat{i}) dy dz$$

$$= \int_{z=0}^2 \int_{y=0}^3 0 dy dz$$

$$= 0$$

$$\begin{aligned} \text{Therefore, } \int_S \vec{F} \cdot d\vec{S} &= S_1 + S_2 + S_3 + S_4 + S_5 + S_6 \\ &= -\frac{9}{2} + \frac{9}{2} + 2 - 2 + 6 + 0 \\ &= 6 \end{aligned}$$

$$\therefore \int_V \text{div} \cdot \vec{F} dv = \int_S \vec{F} \cdot d\vec{S} = 6$$

Hence, Divergence theorem Verified. (Ans.)

Ques: 75 If $\vec{F} = 2z\hat{i} - x\hat{j} + y\hat{k}$; Evaluate $\int_V \vec{F} \cdot d\vec{V}$ where V is bounded by the surfaces $x=0, x=2, y=0, y=4, z=x^2, z=2$

Solution: Given that,

$$\vec{F} = 2z\hat{i} - x\hat{j} + y\hat{k}$$

Also the equation of planes $x=0, x=2, y=0, y=4, z=x^2, z=2$

$$\text{Now, } \int_V \vec{F} \cdot d\vec{V} = \int_{x=0}^2 \int_{y=0}^4 \int_{z=x^2}^2 (2z\hat{i} - x\hat{j} + y\hat{k}) dz dy dx$$

$$= \int_{x=0}^2 \int_{y=0}^4 \left[2z\hat{i} - xz\hat{j} + yz\hat{k} \right]_{z=x^2}^2 dy dx$$

$$= \int_{x=0}^2 \int_{y=0}^4 [2z\hat{i} - xz\hat{j} + yz\hat{k}]_{z=x^2}^2 dy dx$$

$$= \int_{x=0}^2 \int_{y=0}^4 [4\hat{i} - 2xz\hat{j} + 2yz\hat{k} - x^4\hat{i} + x^3\hat{j} - x^2y\hat{k}] dy dx$$

$$= \int_{x=0}^2 \left[4y\hat{i} - 2xy\hat{j} + x \cdot \frac{y^2}{2}\hat{k} - x^4y\hat{i} + x^3y\hat{j} - x^2 \cdot \frac{y^2}{2}\hat{k} \right]_{y=0}^4 dx$$

$$= \int_{x=0}^2 (16\hat{i} - 8x\hat{j} + 16\hat{k} - 4x^4\hat{i} + 4x^3\hat{j} - \frac{x^2}{2} \cdot 16\hat{k}) dx$$

$$= \left[16x\hat{i} - 8 \cdot \frac{x^2}{2}\hat{j} + 16x\hat{k} - 4 \frac{x^5}{5}\hat{i} + 4 \frac{x^4}{4}\hat{j} - 8 \cdot \frac{x^3}{3}\hat{k} \right]_{x=0}^2$$

$$= 32\hat{i} - 16\hat{j} + 32\hat{k} - \frac{128}{5}\hat{i} + 16\hat{j} - \frac{64}{3}\hat{k}$$

$$= \frac{32}{5}\hat{i} + \frac{32}{3}\hat{k}$$

(Ans.)

Ques: 76 Evaluate $\int_V \vec{F} \cdot d\vec{v}$ where V is the region bounded by the planes:

$x=0, x=2, y=0, y=3, z=0, z=4$ and

$$\vec{F} = xy\hat{i} + z\hat{j} - x^2\hat{k}$$

Solution: Given that,

$$\vec{F} = xy\hat{i} + z\hat{j} - x^2\hat{k}$$

Also the equation of planes are,

$x=0, x=2, y=0, y=3, z=0, z=4$

Now,

$$\int_V \vec{F} \cdot d\vec{v} = \iiint_V (xy\hat{i} + z\hat{j} - x^2\hat{k}) \cdot d\vec{v}$$

$$= \int_{x=0}^2 \int_{y=0}^3 \int_{z=0}^4 (xy\hat{i} + z\hat{j} - x^2\hat{k}) dz dy dx$$

$$= \int_{x=0}^2 \int_{y=0}^3 \left[xyz\hat{i} + \frac{z^2}{2}\hat{j} - x^2z\hat{k} \right]_{z=0}^4 dy dx$$

$$= \int_{x=0}^2 \int_{y=0}^3 (4xy\hat{i} + 8\hat{j} - 4x^2\hat{k}) dy dx$$

$$= \int_{x=0}^2 \left[4x \frac{y^2}{2} \hat{i} + 8y\hat{j} - 4x^2 \cdot y\hat{k} \right]_{y=0}^3 dx$$

$$= \int_{x=0}^2 \left(\frac{36x}{2} \hat{i} + 24\hat{j} - 12x^2\hat{k} \right) dx$$

$$= \left[\frac{36}{2} \cdot \frac{x^2}{2} \hat{i} + 24x\hat{j} - 12 \cdot \frac{x^3}{3} \hat{k} \right]_{x=0}^2$$

$$= 36\hat{i} + 48\hat{j} - 32\hat{k}$$

$$= 4(9\hat{i} + 12\hat{j} - 8\hat{k})$$

(Ans.)

Ques: 78 Show that $\iint_S \vec{F} \cdot \hat{n} \, ds = \frac{3}{2}$; where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by the planes $x=0, x=1, y=0, y=1, z=0, z=1$

Solution: Given that,

$$\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$$

Also the equations of planes are,

$$x=0, x=1, y=0, y=1, z=0, z=1$$

We know that,

$$\oint_S \vec{F} \cdot d\vec{s} = \oint_S \vec{F} \cdot \hat{n} \cdot ds$$

Now,

$S_1(OABC)$:

where $z=0, \hat{n} = -\hat{k}$ and $ds_1 = dx dy$

$$\therefore \iint_{S_1} \vec{F} \cdot \hat{n} \, ds_1 = \iint_{S_1} (4xz\hat{i} - y^2\hat{j} + yz\hat{k}) (-\hat{k}) \, ds_1$$

$$= \int_{y=0}^1 \int_{x=0}^1 (-yz) \, dx \, dy$$

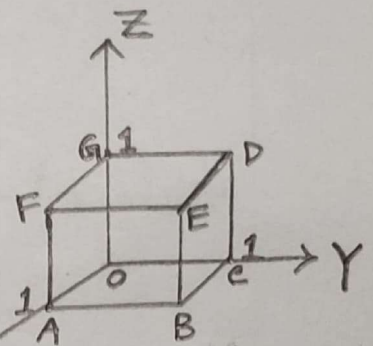
$$= \int_{y=0}^1 [-xyz]_{x=0}^1 \, dy$$

$$= \int_{y=0}^1 (-yz) \, dy$$

$$= \left[-\frac{y^2}{2} \cdot z \right]_{y=0}^1$$

$$= -\frac{1}{2} \cdot z \Rightarrow -\frac{1}{2} \times 0 \quad [\because z=0]$$

$$= 0$$



Also, $S_2(DEF G)$:

where $z=1, \hat{n}=\hat{k}$ and $ds_2 = dx dy$

$$\begin{aligned}\therefore \iint_{S_2} \vec{F} \cdot \hat{n} \cdot ds_2 &= \iint_{S_2} (4xz \cdot 1\hat{i} - y^2\hat{j} + y \cdot 1\hat{k}) \cdot (\hat{k}) \cdot ds_2 \\ &= \int_{y=0}^1 \int_{x=0}^1 (y) dx dy \\ &= \int_{y=0}^1 [xy]_{x=0}^1 dy \\ &= \int_{y=0}^1 y dy \\ &= \left[\frac{y^2}{2} \right]_{y=0}^1 \\ &= \frac{1}{2}\end{aligned}$$

Also, $S_3(AO G F)$:

where, $y=0, \hat{n}=-\hat{j}$ and $ds_3 = dx dz$

$$\begin{aligned}\therefore \iint_{S_3} \vec{F} \cdot \hat{n} \cdot ds_3 &= \iint_{S_3} (4xz \hat{i} - 0^2 \hat{j} + 0 \cdot z \hat{k}) \cdot (-\hat{j}) \cdot ds_3 \\ &= \int_{z=0}^1 \int_{x=0}^1 0 dx dz \\ &= 0\end{aligned}$$

Also, $S_4(BCDE)$:

where, $y=1, \hat{n}=\hat{j}$ and $ds_4 = dx dz$

$$\begin{aligned}\therefore \iint_{S_4} \vec{F} \cdot \hat{n} \cdot ds_4 &= \iint_{S_4} (4xz \hat{i} - 1\hat{j} + z \hat{k}) \cdot (\hat{j}) \cdot ds_4 \\ &= \int_{z=0}^1 \int_{x=0}^1 (-1) dx dz\end{aligned}$$

$$= \int_{z=0}^1 [-x]_{x=0}^1 dz$$

$$= \int_{z=0}^1 -1 dz$$

$$= [-z]_{z=0}^1$$

$$= -1$$

Also, S_5 (OCDG):

Where, $x=0$, $\hat{n} = -\hat{i}$ and $ds_5 = dy dz$

$$\begin{aligned} \therefore \iint_{S_5} \vec{F} \cdot \hat{n} ds_5 &= \iint_{S_5} (4xz\hat{i} - y^2\hat{j} + yz\hat{k}) (-\hat{i}) \cdot ds_5 \\ &= \int_{z=0}^1 \int_{y=0}^1 0 \cdot dy dz \\ &= 0 \end{aligned}$$

Also, S_6 (ABEF):

Where $x=1$, $\hat{n} = \hat{i}$ and $ds_6 = dy dz$

$$\begin{aligned} \therefore \iint_{S_6} \vec{F} \cdot \hat{n} ds_6 &= \iint_{S_6} (4 \cdot 1 \cdot z\hat{i} - y^2\hat{j} + yz\hat{k}) (\hat{i}) ds_6 \\ &= \int_{z=0}^1 \int_{y=0}^1 (4z) dy dz \\ &= \int_{z=0}^1 [4yz]_{y=0}^1 dz \\ &= \int_{z=0}^1 4z dz \end{aligned}$$

$$= \left[\frac{4z^2}{2} \right]_{z=0}^1$$

$$= 2 \times 1$$

$$= 2$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = S_1 + S_2 + S_3 + S_4 + S_5 + S_6$$

$$= 0 + \frac{1}{2} + 0 + (-1) + 0 + 2$$

$$= \frac{1}{2} - 1 + 2$$

$$= \frac{3}{2}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \frac{3}{2} \quad [\text{Showed}]$$

$$= 0$$

$$= 0$$

$$= 0$$

$$= 0$$

$$= 0$$

$$= 0$$

$$= 0$$

$$= 0$$